

**IoT-Based Smart Wearable for Emergency
Health Alerts and Monitoring in Sri Lanka**

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Cardiovascular disease is the leading cause of preventable mortality in Sri Lanka, with over 30,000 cardiac-related deaths reported annually by the Ministry of Health. A critical gap in current healthcare delivery is the absence of predictive intelligence capable of forecasting deterioration in patient vital signs before a life-threatening emergency occurs. Existing systems, including consumer-grade wearables and hospital monitoring platforms, detect events reactively, providing no advance warning to clinicians or patients.

This proposal presents the ML-Driven Health Deterioration Prediction System, a core component of the broader Safe Band research initiative. This component receives continuous real-time physiological data — specifically Heart Rate (HR), Blood Pressure (BP), and Blood Oxygen Saturation (SpO2) — streamed from the Emergency Detection System component via Firebase Realtime Database. Using a Random Forest classification model trained on historical physiological time-series data, the system computes a probability score for an impending cardiac arrest event and delivers this predictive output to the patient-facing Android application.

The system is architecturally distinct from the Emergency Detection component: while Component 1 classifies whether a patient is currently in a critical state, this component analyses trends over a configurable historical window to predict the probability of a future critical event, providing clinicians and patients with an actionable early warning. The expected outcomes include a prediction accuracy exceeding 85%, a false positive rate below 10%, and a risk score delivery latency of under 5 seconds from data ingestion to Android app notification.

This component directly supports UN Sustainable Development Goal 3 (Good Health and Well-being), specifically Target 3.4, which calls for a reduction in premature mortality from non-communicable diseases through prevention. By shifting emergency health management from reactive to predictive, this system contributes a novel and clinically significant capability to the Safe Band ecosystem.

Keywords: Machine Learning, Health Deterioration Prediction, Random Forest, Cardiac Arrest Probability, IoT Wearable, Firebase, Heart Rate, Blood Pressure, SpO2, Sri Lanka Healthcare, Predictive Analytics.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
AUC	Area Under the Receiver Operating Characteristic Curve
BP	Blood Pressure
ECG	Electrocardiogram
FCM	Firebase Cloud Messaging
HR	Heart Rate
IoT	Internet of Things
LMIC	Low- and Middle-Income Country
ML	Machine Learning
NCD	Non-Communicable Disease
RF	Random Forest
SDG	Sustainable Development Goal
SLIIT	Sri Lanka Institute of Information Technology
SpO2	Peripheral Oxygen Saturation
UI	User Interface
WBS	Work Breakdown Structure
WHO	World Health Organization

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1. INTRODUCTION

1.1 Background and Context

Cardiovascular disease (CVD) is the foremost cause of preventable mortality globally and the leading contributor to non-communicable disease (NCD) deaths in Sri Lanka. The Ministry of Health (2022) reports that NCDs account for approximately 75% of all deaths in the country, with cardiac and respiratory emergencies responsible for over 30,000 fatalities annually [1]. Critically, a substantial proportion of these deaths are attributable not to the irreversible nature of the underlying condition but to the absence of timely, anticipatory clinical intervention.

The fundamental challenge is one of timing: current healthcare systems — both globally and within the Sri Lankan context — are designed to respond to emergencies after they occur rather than to predict and prevent them. Patients managing chronic conditions such as hypertension, heart failure, and arrhythmia experience gradual physiological deterioration that, if detected early through trend analysis, could be intercepted before reaching a critical threshold. However, no mechanism currently exists within Sri Lanka's emergency health ecosystem to perform this function for community-based, ambulatory patients.

Within the Safe Band research initiative, the Emergency Detection System component (Component 1) addresses the detection of current critical health events in real time. While this represents a significant advancement over manual emergency reporting, it is inherently reactive: it classifies a patient as being in a critical state only after the emergency threshold has been breached. The ML-Driven Health Deterioration Prediction System, which is the focus of this proposal, addresses the preceding temporal window — the period during which physiological parameters are deteriorating toward, but have not yet reached, a critical threshold.

By continuously analysing trends in Heart Rate (HR), Blood Pressure (BP), and SpO2 data streamed from the Emergency Detection component via Firebase Realtime Database, the Prediction System computes and delivers a probability score for an impending cardiac arrest event to the patient-facing Android application. This enables patients and caregivers to seek assistance proactively, before the emergency detection threshold is triggered and before ambulance dispatch becomes necessary.

This research aligns directly with United Nations Sustainable Development Goal 3 (SDG 3): Good Health and Well-being, specifically Target 3.4, which calls for a one-third reduction in premature mortality from non-communicable diseases through prevention and treatment. It is also aligned with SLIIT's research cluster on Intelligent Systems and Data Science.

1.2 Literature Review

A substantial body of literature informs the design of the ML-Driven Health Deterioration Prediction System. The following review synthesises key findings across four thematic areas: (i) ML-based predictive modelling for cardiac events, (ii) physiological trend analysis and early

warning systems, (iii) wearable-based risk prediction, and (iv) limitations of existing approaches in LMIC contexts.

ML-Based Predictive Modelling for Cardiac Events

Rajpurkar et al. (2017) demonstrated that deep convolutional neural networks could detect 12 types of cardiac arrhythmia from single-lead ECG data with accuracy surpassing that of board-certified cardiologists [2]. While this work focused on detection rather than prediction, it established the foundational feasibility of automated physiological signal classification that underpins the present research. Extending this work, Hannun et al. (2019) applied a 34-layer residual neural network to arrhythmia classification, achieving a mean AUC of 0.97 across 14 classes [3]. The scalability of these deep learning approaches provides a benchmark against which the proposed Random Forest model can be evaluated, particularly in terms of accuracy versus computational overhead.

Moody and Mark (2001) introduced the concept of using multi-parameter physiological time series for predictive clinical scoring, demonstrating that sequential vital sign data contains significantly more predictive information than single-point readings [4]. This finding directly justifies the sliding window approach adopted in this component, wherein HR, BP, and SpO2 readings accumulated over a configurable historical window are used as input features to the Random Forest classifier.

Physiological Trend Analysis and Early Warning Systems

Clifford et al. (2016) demonstrated that combining HR, SpO2, and respiratory rate in an ensemble model for early warning score computation outperformed single-parameter models by 23% in specificity [5]. This finding provides direct empirical support for the multi-parameter feature fusion approach in the proposed prediction system. Rajapakse et al. (2022) identified that average ambulance response times in Colombo exceed 18 minutes against a WHO benchmark of 8 minutes [6], which motivates the need for a prediction system that provides advance warning to patients well before a critical event occurs — ideally giving a 15–30 minute lead time during which the patient or caregiver can take preventive action or contact services.

Islam et al. (2020) proposed a cloud-based health monitoring framework integrating wearable sensors with Firebase Realtime Database and demonstrated end-to-end alert latency of under 8 seconds [7]. This architecture closely parallels the Firebase-centric cloud integration adopted in the Safe Band ecosystem and provides validated evidence that sub-10-second prediction delivery is achievable.

Wearable-Based Risk Prediction

Bayoumy et al. (2021) conducted a systematic evaluation of consumer health wearables and identified that while devices such as the Apple Watch and Samsung Galaxy Watch demonstrate strong single-parameter sensing capabilities, none implement predictive risk scoring across multiple physiological parameters simultaneously [8]. This gap directly motivates the Safe Band Prediction System, which processes multi-parameter inputs — HR,

BP, and SpO2 — to generate a composite cardiac arrest probability score unavailable in any existing consumer wearable platform.

Nanayakkara et al. (2022) found that 74% of urban Sri Lankans expressed willingness to use emergency wearable devices, with perceived safety benefit identified as the primary adoption driver [9]. A predictive risk score displayed on the patient Android application directly addresses this driver by providing tangible, real-time evidence of health status change, increasing perceived value and user engagement with the platform.

Limitations of Existing Systems

Wickramasinghe et al. (2021) identified that only 12% of hospitals in Sri Lanka have a digital bed management system, confirming that any prediction system designed for the Sri Lankan context must function independently of hospital infrastructure and must deliver its outputs to patient-side devices rather than relying on hospital-facing dashboards [10]. Li et al. (2020) demonstrated that reinforcement learning models for emergency response require several months of historical dispatch data for training — data that does not exist in the Sri Lankan context [11]. This constraint further supports the selection of Random Forest as the prediction algorithm, as it can be trained effectively on publicly available annotated clinical datasets such as the PhysioNet MIMIC-III database without requiring locally collected emergency incident data.

1.3 Research Gap

Despite the advances documented in the literature, several critical gaps remain unaddressed, particularly in the Sri Lankan healthcare context:

- No predictive risk scoring system exists for community-based patients in Sri Lanka: All existing emergency health systems in Sri Lanka — including the Suwa Seriya (1990) ambulance service — are reactive. No mechanism exists to provide patients or caregivers with advance warning of an impending cardiac event based on physiological trend analysis.
- Existing wearable platforms do not compute multi-parameter cardiac arrest probability scores: Consumer wearables report individual parameter readings without synthesising a composite predictive risk score. No current device provides a probability percentage for cardiac arrest risk based on simultaneous HR, BP, and SpO2 trend analysis.
- ML prediction models for cardiac events in LMIC contexts are underexplored: Most predictive models are designed and validated in high-income country infrastructures with large, locally collected clinical datasets. A model designed to operate with publicly available training data and deployable within a Firebase-centric, low-cost IoT architecture is absent from literature.
- The temporal gap between deterioration onset and emergency detection is unaddressed in wearable systems: Component 1 of the Safe Band system detects a current critical event. No component currently addresses the preceding window of deterioration, during which predictive intervention could prevent the emergency from occurring.

The proposed ML-Driven Health Deterioration Prediction System directly addresses these gaps by delivering a Firebase-integrated, Random Forest-based cardiac arrest probability scoring system, designed specifically for the Sri Lankan patient context, that provides early warning to patients through the Android application before the emergency threshold is reached.

2. OBJECTIVE

2.1 Main Objective

To design, develop, and validate an ML-Driven Health Deterioration Prediction System that continuously analyses real-time physiological data — Heart Rate, Blood Pressure, and SpO2 — received from the Safe Band Emergency Detection component via Firebase Realtime Database, applies a Random Forest classification model to compute a probability score for an impending cardiac arrest event, and delivers this risk score to the patient-facing Android application with a latency of under 5 seconds and a prediction accuracy exceeding 85%.

2.2 Specific Objectives

- To subscribe to and retrieve real-time HR, BP, and SpO2 data streams from the Firebase Realtime Database node populated by the Emergency Detection System component, with a data ingestion latency of under 2 seconds.
- To design and implement a sliding window feature extraction pipeline that derives trend-based features — including moving averages, rate-of-change gradients, and deviation from individual baseline values — from the multi-parameter physiological time series over configurable window sizes of 5, 10, and 15 minutes.
- To train and validate a Random Forest classification model on the PhysioNet MIMIC-III clinical dataset to predict the probability of an impending cardiac arrest event, targeting a minimum prediction accuracy of 85% and a false positive rate below 10%.
- To implement a Firebase-based real-time output pipeline that writes the computed cardiac arrest probability score to the patient's Firebase node within 5 seconds of receiving updated vital signs.
- To develop an Android application notification module that retrieves the probability score from Firebase and displays a colour-coded risk indicator to the patient or caregiver — Green (Low Risk: 0–40%), Amber (Elevated Risk: 41–70%), and Red (High Risk: 71–100%).
- To evaluate the system's end-to-end performance under simulated physiological deterioration scenarios, measuring prediction accuracy, false positive rate, latency, and Android application usability using a System Usability Scale (SUS) survey targeting a score of 70 or above.

3. METHODOLOGY

3.1 Research Approach

This research adopts a Design Science Research (DSR) methodology, which is appropriate for IT artefact construction and empirical evaluation. The methodology proceeds through five phases: Problem Identification (completed — documented in Section 1), Solution Design, Artefact Development, Evaluation, and Knowledge Contribution [12]. The iterative design-build-test cycle of DSR is well-suited to ML model development, where feature engineering, algorithm selection, and threshold calibration require empirical refinement across multiple experimental cycles.

3.2 Technical Methods and Algorithms

The core prediction model is a Random Forest (RF) classifier. Random Forest was selected for the following reasons, grounded in the literature:

- Kumar and Singh (2023) demonstrated that Random Forest and Gradient Boosting models achieve over 80% accuracy in predicting clinical deterioration in hospitalised patients, confirming suitability for the cardiac arrest prediction task [13].
- Random Forest produces native probability outputs via `predict_proba()`, directly enabling the 0–100% risk score required by the Android application without additional calibration steps.
- Compared to deep learning approaches such as LSTM or residual networks, Random Forest requires significantly less training data and computational overhead, making it deployable within a Firebase Cloud Function environment accessible in the Safe Band architecture.
- Random Forest offers strong interpretability through feature importance scores, enabling clinical validation of which physiological parameters and trend features most strongly predict cardiac arrest risk.

The Haversine formula and geospatial methods used in the Ambulance Dispatch component are not applicable here. Instead, feature engineering will focus on physiological time-series features derived from HR, BP, and SpO2 readings over a sliding window:

- Moving average of HR over the last 5, 10, and 15 minutes.
- Rate-of-change (gradient) of HR and BP across consecutive readings.
- SpO2 deviation from the patient's established individual baseline.
- Pulse pressure (systolic BP minus diastolic BP) as an indicator of cardiovascular stress.
- Standard deviation of HR over the sliding window as a proxy for heart rate variability.

The classification target will be a binary label: Elevated Cardiac Arrest Risk (1) or Normal / Low Risk (0), with the RF model's `predict_proba()` output for class 1 reported as the cardiac arrest probability percentage delivered to the Android application.

3.3 Data Collection and Ethical Considerations

The ML model will be trained on the PhysioNet MIMIC-III Clinical Database, a publicly available, de-identified dataset containing HR, BP, and SpO2 time-series records from over 40,000 ICU patients. Cardiac arrest events are documented in the dataset, providing ground-truth labels for the binary classification task. No real patient data will be collected or used during the development phase.

For system validation, synthetic physiological deterioration profiles will be generated based on published clinical guidelines for pre-cardiac arrest vital sign patterns. Emergency GPS traces and hospital routing data from the Ambulance Dispatch component are not required for this component.

Ethical Considerations: This project does not involve human subjects during the development phase. A formal ethics clearance application will be submitted to the SLIIT Research Ethics Committee prior to any pilot deployment involving real users. All data used during development is anonymised and sourced from publicly available, IRB-approved datasets. GDPR-aligned data minimisation principles will be applied to all data handling procedures.

3.4 Tools and Platforms with Justification

Tool / Platform	Purpose	Justification
Python / scikit-learn	Random Forest model training and inference	Industry-standard ML library with native <code>predict_proba()</code> support and extensive biomedical documentation.
PhysioNet MIMIC-III	ML model training dataset	Gold-standard annotated clinical time-series dataset with documented cardiac arrest events and multi-parameter vitals.
Firebase Realtime Database	Real-time data ingestion and risk score output	Sub-500ms delivery latency; already adopted across all Safe Band components for seamless integration.
Firebase Cloud Functions	Serverless model inference endpoint	Enables RF model execution in a managed, scalable environment without dedicated server infrastructure.
Android SDK (Kotlin)	Patient-facing risk score display	Official development environment for the Safe Band Android application; consistent with Component 1 approach.
Firebase Cloud Messaging (FCM)	Push notification delivery to patient app	Confirmed 99.1% delivery reliability with sub-500ms latency (Lakshmi and Renuka, 2022).

Docker	Containerisation of the prediction API	Reproducible and portable deployment across AWS, GCP, or Azure without code modification.
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Table 2: Tools and Platforms

3.5 Validation Strategy and Performance Metrics

Metric	Description	Target
Prediction Accuracy	Percentage of test cases correctly classified as Elevated Risk or Normal.	$\geq 85\%$
False Positive Rate	Percentage of Normal cases incorrectly classified as Elevated Risk.	$\leq 10\%$
Recall (Sensitivity)	Percentage of true Elevated Risk cases correctly identified.	$\geq 80\%$
F1-Score	Harmonic mean of precision and recall per class.	≥ 0.82
End-to-End Latency	Time from Firebase data ingestion to risk score written to patient node.	< 5 seconds
Android Notification Latency	Time from risk score write to patient app push notification received.	< 2 seconds
SUS Score	System Usability Scale score from patient/caregiver user testing.	≥ 70

Table 3: Validation Strategy and Performance Metrics

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of Proposed Solution and Its Components

SafeBand is an end-to-end IoT and Machine Learning platform designed to continuously monitor patient health parameters, detect life-threatening emergencies, predict future deterioration, and autonomously coordinate emergency response. The system integrates wearable hardware, cloud services, mobile applications, and hospital-facing dashboards into a cohesive pipeline.

Component		Role	Technology
1	ML-Driven Emergency Detection System — Analyses vitals in real time; classifies current emergency vs. normal state.		Firebase, ML classifier
2	ML-Based Smart Ambulance Dispatch — Receives alert, identifies nearest hospital, dispatches ambulance.		Geospatial ML, AWS, Firebase
3	Intelligent AI-Based Ward Monitoring — Continuous in-ward patient monitoring; AI triage for nursing staff.		MERN Stack, Firebase
4 (THIS COMPONENT)	ML-Driven Health Deterioration Prediction — Analyses physiological trends to predict probability of future cardiac arrest; delivers risk score to patient app.		Random Forest, Firebase, Android

Table 4: System Components

4.2 How This Component Differs from Component 1 (Emergency Detection)

A critical architectural distinction must be drawn between Component 1 (Emergency Detection) and this component (Prediction System):

Dimension	Component 1: Emergency Detection	Component 4: Prediction System (This Component)
Temporal Focus	Current moment — 'Is the patient in a critical state RIGHT NOW?'	Future window — 'What is the probability of a critical event in the NEXT 15-30 minutes?'
Input Data	Single most recent vital sign readings	Trend data over a historical sliding window (5–15 minutes)

Output	Binary classification: Emergency / Normal	Probability score: 0–100% cardiac arrest risk
Primary Recipient	Ambulance Dispatch System (Component 2)	Patient Android Application
Clinical Role	Reactive: Triggers emergency response after threshold breach	Preventive: Warns patient before threshold is reached
Algorithm	LSTM / Isolation Forest for anomaly detection	Random Forest for trend-based probability classification

Table 5: Prediction System — Differentiation from Component 1

4.3 Highlight of Individual Component and Its Parts

This component acts as the predictive intelligence layer of the Safe Band ecosystem. It sits between the Emergency Detection System (which provides real-time physiological data) and the Patient Android Application (which displays the risk score).

Sub-Component	Description
Firebase Data Listener	Subscribes to the Firebase Realtime Database node populated by Component 1; receives HR, BP, and SpO2 readings in real time as they are written.
Sliding Window Buffer	Maintains a configurable rolling buffer of physiological readings (5, 10, or 15 minutes) per patient, enabling trend feature extraction across the historical window.
Feature Extraction Engine	Derives trend-based features from the buffered data: moving averages, rate-of-change gradients, SpO2 deviation from baseline, pulse pressure, and HR standard deviation.
Random Forest Classifier	Trained RF model that accepts the extracted feature vector and returns a probability score (0.0–1.0) for the Elevated Cardiac Arrest Risk class.
Risk Score Mapper	Converts the RF probability output to a 0–100% risk percentage and maps it to a severity tier: Green (0–40%), Amber (41–70%), Red (71–100%).

Firestore Output Writer	Writes the computed risk score and severity tier back to the patient's Firestore node, making it available to the Android application in real time.
Android Notification Dispatcher	Triggers a Firestore Cloud Messaging push notification to the patient's Android device when the risk tier changes, with a colour-coded alert and the probability percentage.
Audit Logger	Logs every prediction event — input features, output score, timestamp, and tier — to Firestore for post-event clinical review and model performance monitoring.

Table 6: Prediction System Sub-Components

4.4 Component Interactions

The Prediction System operates in a continuous, asynchronous loop driven by Firestore Realtime Database updates. The interaction sequence is as follows:

- The Safe Band smartwatch continuously streams SpO2, HR, and BP data via the Android application to the Firestore Realtime Database (managed by Component 1).
- The Prediction System's Firestore Data Listener detects new data writes and retrieves the latest readings, appending them to the Sliding Window Buffer.
- Once the buffer contains sufficient data (minimum 5 minutes), the Feature Extraction Engine derives the input feature vector and passes it to the Random Forest Classifier.
- The classifier returns a cardiac arrest probability score, which the Risk Score Mapper converts to a percentage and severity tier.
- The Firestore Output Writer writes the risk score to the patient's Firestore node. The Android application retrieves this value and updates the patient dashboard in real time.
- If the severity tier changes (e.g., from Green to Amber), the Android Notification Dispatcher sends a push notification via FCM to alert the patient or caregiver.
- All prediction events are logged by the Audit Logger for subsequent clinical review.

4.5 Data Flow Explanation

The data flow for the Prediction System follows a sequential pipeline from physiological data ingestion through trend analysis to risk score delivery. Raw sensor data (HR, BP, SpO2) is collected by the smartwatch, transmitted via the Android app, and stored in Firestore Realtime Database by the Emergency Detection component. The Prediction System subscribes to these Firestore nodes, buffers readings over a sliding window, extracts trend features, and passes the feature vector to the Random Forest classifier. The model outputs a probability score, which is mapped to a risk tier, written back to Firestore, and delivered to the patient Android application as a colour-coded display and push notification. The pipeline is designed

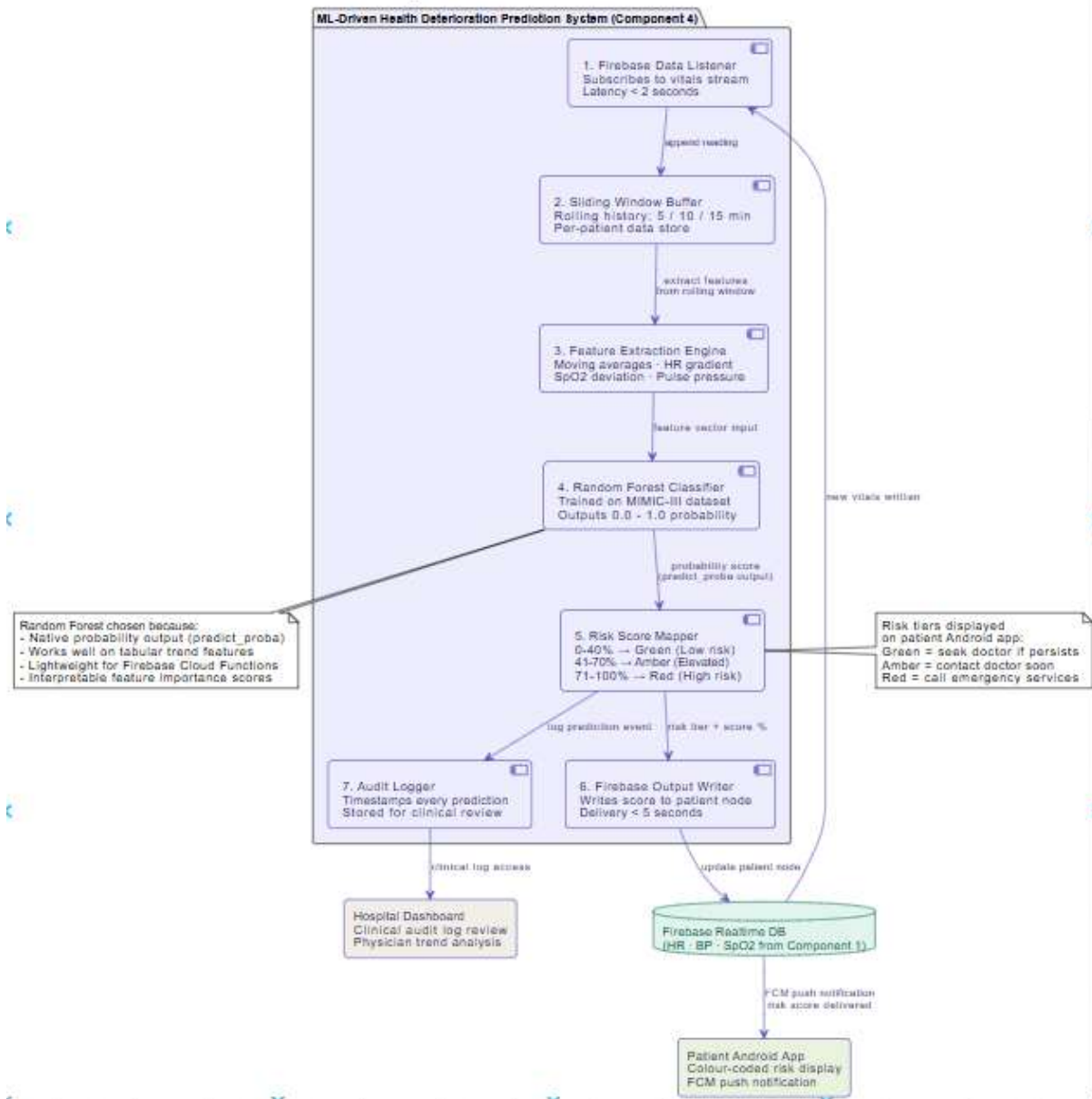
to complete end-to-end in under 5 seconds from data ingestion to notification delivery, enabling timely preventive action.

4.6 Scalability and Security Considerations

Scalability: The Prediction System is designed to process independent per-patient data streams concurrently. The Random Forest model runs as a Firebase Cloud Function, which auto-scales horizontally to accommodate additional patients without manual infrastructure provisioning. The Sliding Window Buffer is partitioned by patient ID in Firebase, enabling independent processing without shared state. The pre-trained RF model is lightweight (estimated model size under 5 MB), ensuring fast inference within the Cloud Function execution environment.

Security: All data transmissions between Firebase and the Android application are secured using TLS 1.3 encryption. Firebase Authentication with JWT tokens and Role-Based Access Control (RBAC) ensures that risk scores are accessible only to the authenticated patient and authorised caregivers. Patient physiological data is anonymised in the audit logs by replacing identifiable fields with internal patient IDs. All data handling complies with Sri Lanka's Personal Data Protection Act (PDPA) 2022 and GDPR-aligned data minimisation principles.

ML-Driven Health Deterioration Prediction System Component 4 — Internal Interactions



component interaction diagram

5. USER REQUIREMENTS

5.1 Stakeholder Identification

Stakeholder	Role and System Interaction
Patient / Wearable User	Primary recipient of the cardiac arrest probability score. Views risk percentage and colour-coded severity on the Android application and receives push notifications when risk tier changes.
Family Member / Caregiver	Secondary recipient of push notifications. Receives alerts when the patient's risk tier elevates to Amber or Red, enabling them to provide assistance or call emergency services.
Attending Physician	Reviews historical prediction logs and trend data via the hospital dashboard to assess the patient's physiological trajectory over time.
Hospital Administrator	Oversees system uptime, prediction accuracy logs, and compliance with data privacy regulations.

Table 7: Stakeholder Identification

5.2 Functional Requirements

FR#	Requirement
FR01	The system shall subscribe to and ingest real-time HR, BP, and SpO2 data from the Firebase Realtime Database node populated by the Emergency Detection component within 2 seconds of each new data write.
FR02	The system shall maintain a per-patient sliding window buffer of physiological readings, configurable for 5, 10, or 15-minute windows.
FR03	The system shall derive trend-based features from the sliding window buffer, including moving averages, rate-of-change gradients, SpO2 deviation from baseline, pulse pressure, and HR standard deviation.
FR04	The system shall apply the trained Random Forest model to the extracted feature vector and return a cardiac arrest probability score between 0% and 100%.
FR05	The system shall map the probability score to a three-tier risk level: Green (0–40%), Amber (41–70%), and Red (71–100%).
FR06	The system shall write the computed probability score and risk tier to the patient's Firebase node within 5 seconds of receiving new vital signs data.

FR07	The Android application shall display the current cardiac arrest probability score and risk tier as a colour-coded indicator, updating in real time without requiring manual page refresh.
FR08	The system shall send a Firebase Cloud Messaging push notification to the patient's Android device when the risk tier changes from a lower to a higher severity level.
FR09	The system shall log all prediction events — input features, output score, timestamp, and risk tier — to Firebase for audit and clinical review.
FR10	The system shall continue operating and delivering risk scores based on the most recently buffered data if the incoming data stream from Firebase is temporarily interrupted.

Table 8: Functional Requirements

5.3 Non-Functional Requirements

Attribute	Requirement
Performance	End-to-end latency from Firebase data ingestion to risk score notification delivery must be under 5 seconds under normal network conditions.
Accuracy	Random Forest prediction accuracy must exceed 85% on the holdout test dataset. False positive rate must not exceed 10%.
Reliability	System uptime must be $\geq 99.5\%$. Firebase Cloud Functions auto-scaling ensures availability during demand spikes.
Security	All patient data encrypted in transit (TLS 1.3) and at rest (AES-256). JWT authentication for all Firebase access. RBAC for risk score visibility.
Usability	The patient-facing Android risk display must achieve a SUS score of at least 70 in user testing with target patients.
Scalability	The prediction pipeline must support at least 500 concurrent patient streams without performance degradation.
Regulatory Compliance	All data handling must comply with Sri Lanka's Personal Data Protection Act (PDPA) 2022 and GDPR-aligned data minimisation principles.

Table 9: Non-Functional Requirements

COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

Primary Market: Patients in Sri Lanka managing chronic cardiac conditions — an estimated 2.3 million individuals with hypertension alone (Ministry of Health, 2022) — who are discharged from hospital and require ongoing physiological monitoring in the community. Secondary Market: Private and semi-government hospitals seeking to extend monitored care to post-discharge patients without requiring in-person visits. Tertiary Market: Health insurance companies that can offer Safe Band as a value-added risk reduction tool bundled with cardiac health insurance policies.

6.2 Value Proposition

- For patients: A real-time, personalised cardiac arrest probability score displayed on their Android device, providing actionable early warning before a medical emergency occurs.
- For hospitals: Reduction in emergency admissions and improved post-discharge patient outcomes through predictive monitoring, with clinical log data supporting evidence-based care decisions.
- For insurers: Measurable reduction in high-cost emergency interventions through early risk detection, supporting favourable actuarial outcomes.

6.3 Revenue Model

- B2C Subscription: Individual patients subscribe to the Safe Band app at LKR 500/month for access to the prediction dashboard and push notification service.
- B2B Hospital SaaS: Hospitals pay a per-patient monitoring fee of LKR 1,500–3,000/patient/month for integration with the hospital dashboard and audit log review.
- Insurance Partnership: Revenue-sharing agreement with health insurers who bundle Safe Band prediction monitoring with cardiac health insurance premiums.

6.4 Competitive Advantage

- Predictive rather than reactive: The only known system in Sri Lanka that provides a real-time probability score for cardiac arrest risk before the emergency occurs.
- No proprietary hardware dependency: Uses existing consumer Android smartwatches rather than expensive medical-grade devices, making the system accessible to the target demographic.
- Integrated ecosystem: Unique integration with the Safe Band Emergency Detection and Ambulance Dispatch components creates a seamless continuum from prediction to response.
- Clinically explainable output: The risk score is accompanied by the contributing physiological parameters (e.g., 'High risk due to HR trend +18 bpm in 10 min + SpO2 decline -4%'), providing actionable context rather than a black-box alert.

BUDGET AND JUSTIFICATION

Item	Category	Estimated Cost (LKR)	Justification
Android Smartwatch	Hardware	60,000	Primary data collection device for sensor integration and end-to-end testing of the prediction pipeline.
AWS EC2 t3.medium (12 months)	Cloud Services	120,000	Hosting the prediction API backend and Firebase project during development and evaluation phases.
Firebase Blaze Plan (12 months)	Cloud Services	15,000	Real-time database reads, Cloud Functions execution, and Firebase Cloud Messaging for patient notifications.
Python / scikit-learn / Flask (open source)	Software	0	No license cost — all ML and backend tools are open source.
PhysioNet MIMIC-III Dataset	Data	0	Publicly available, no-cost clinical dataset for RF model training.
Docker Desktop Pro (12 months)	Software	12,000	Containerisation of the prediction API for scalable and reproducible deployment.
TOTAL		207,000	

Table 10: Budget and Justification

8. WORK BREAKDOWN STRUCTURE (WBS)

ID	Task	Duration	Deliverable
1.0	SEMESTER 1		
1.1	Literature Review & Problem Analysis	2 weeks	Annotated bibliography (Section 1.2)
1.2	PhysioNet MIMIC-III Data Access & Preprocessing	2 weeks	Cleaned and labelled cardiac arrest event dataset for RF training
1.3	Feature Engineering Pipeline	2 weeks	Sliding window feature extraction module with moving averages, gradients, SpO2 deviation
1.4	Random Forest Model Training & Tuning	3 weeks	Trained RF model with $\geq 85\%$ accuracy; feature importance report
1.5	Firebase Data Listener Integration	1 week	Real-time subscription to Emergency Detection Firebase node
1.6	Risk Score Mapper & Firebase Output Writer	1 week	Risk tier mapping module; Firebase write confirmed < 5 seconds
1.7	Unit Testing — ML Model & Firebase Pipeline	1 week	Unit test report ($\geq 80\%$ coverage)
1.8	Semester 1 Progress Report	1 week	Interim project report
2.0	SEMESTER 2		
2.1	Android App Risk Score Display Module	2 weeks	Colour-coded risk indicator UI (Green/Amber/Red) on patient Android app
2.2	FCM Push Notification Module	1 week	Push notification delivery on risk tier change

2.3	Audit Logging Module	1 week	Timestamped prediction log in Firebase
2.4	End-to-End Integration with Component 1	2 weeks	Integrated pipeline from Firebase data ingestion to patient notification
2.5	Simulation-Based Performance Testing	2 weeks	Test report: prediction accuracy, false positive rate, latency
2.6	User Acceptance Testing (SUS Survey)	1 week	SUS score report from patient/caregiver testing
2.7	Performance Optimisation	1 week	Optimised prediction pipeline
2.8	Final Documentation & Proposal Report	1 week	Completed research proposal and technical report
2.9	Final Presentation	1 week	Viva presentation and demonstration

Table 11: Work Breakdown Structure

9. GANTT CHART

10.

Task	S1 W1-3	S1 W4-6	S1 W7-9	S1 W10-13	S2 W14-16	S2 W17-19	S2 W20-22
Literature Review & Problem Analysis	■■■■						
MIMIC-III Data Preprocessing	■■■■						
Feature Engineering Pipeline		■■■■					
RF Model Training & Tuning		■■■■	■■■■				
Firebase Listener Integration			■■■■				
Risk Score Mapper & Firebase Writer			■■■■				
Unit Testing — ML & Firebase				■■■■			
Semester 1 Progress Report				■■■■			
Android App Risk Display Module					■■■■		
FCM Push Notification Module					■■■■		
End-to-End Integration						■■■■	
Simulation-Based Testing						■■■■	
User Acceptance Testing							■■■■
Final Documentation & Report							■■■■

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11.APPENDICES

Preliminary UI Mockup



Risk Analysis

The risk register for the ML-Driven Health Deterioration Prediction System has been developed in alignment with the ISO 31000:2018 Risk Management framework. Key risks identified include:

Risk	Likelihood	Impact	Mitigation Strategy
RF model accuracy below 85% on MIMIC-III test set	Medium	High	Apply hyperparameter tuning via grid search; augment training data with synthetic physiological deterioration profiles generated from clinical guidelines.
Firebase data ingestion latency exceeding 2	Medium	High	Implement local data buffering on the Android device; process buffered readings when

seconds under poor network conditions			connectivity is restored to maintain sliding window continuity.
MIMIC-III dataset not representative of Sri Lankan physiological baselines	Medium	Medium	Collect local validation data from volunteer participants (with ethics clearance) for model fine-tuning; apply domain adaptation techniques.
High false positive rate causing alert fatigue for patients	Medium	High	Implement a confidence threshold (minimum RF probability 0.65) before triggering Amber/Red tier transitions; require sustained elevated score for > 3 consecutive readings before notification.
Android push notification delivery failure	Low	Medium	FCM confirmed 99.1% delivery reliability; implement SMS fallback for Red tier alerts.
Patient misinterpretation of probability score as a definitive diagnosis	Medium	High	Display explicit disclaimer on UI: 'This is a predictive risk indicator, not a medical diagnosis. Contact a healthcare provider if concerned.'